Ritesh Chowdhry

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EDUCATION

Ph.D. in Electrical and Computer Engineering **Aug. 2021 - Dec. 2026**
University of Florida, Advisor: Dr. Alina Zare *Defended Oct. 2026*
Dissertation: Beyond Band Masking: Decomposition-Guided Self-Supervised Learning for Hyperspectral Images

M.S. in Electrical and Computer Engineering **Aug. 2019 - Aug. 2021**
University of Florida, GPA: 3.8/4.0

SKILLS

- **ML:** LLM Evaluation, Vision-Language Models, Agentic Systems, Self-Supervised Learning (ViT/MAE), Computer Vision
- **Frameworks & Tools:** PyTorch, Lightning, TensorFlow, Hugging Face, AWS Bedrock, GCP, Slurm/HPC, Comet ML
- **Languages & Libraries:** Python, C++, SQL, scikit-learn

WORK EXPERIENCE

Amazon, ***Applied Scientist Intern, Seattle, WA*** **June 2026 - Aug. 2026**
- Built **LLM** routing benchmark labeling correctness, dollar cost, and served latency in the same query-model cell (402,510 cells, 15 models, 11 vendors, 18 datasets, including closed commercial flagships); cheapest-correct and fastest-correct models disagree on 34.1% of solvable queries.
- Developed query-level router pairing a **kNN** capability estimate with a learned head under a joint cost-latency penalty, matching the best single model's accuracy while saving 20.5% cost and 7.9% latency on all 20 seeds; ported six published routers for like-for-like comparison.
- Added **calibrated** confidence gating (calibration error 0.017) lifting accuracy from 77.9% to 89.1% at 80% coverage.

Amazon, ***Applied Scientist Intern, Seattle, WA*** **June 2025 - Aug. 2025**
- Developed agentic system using **VLMs** via **AWS** Bedrock to automate financial report analysis over plots, tables, and markdown, cutting manual work by 20-30 min/report across hundreds of daily scenarios.

NumFOCUS Organization, ***Google Summer of Code Contributor*** **June 2025 - Aug. 2025**
- Developed open-source package NeonCrops with end-to-end NEON data pipeline, multimodal dataset (234 species, 41,738 crowns, 38 sites, RGB/hyperspectral/LiDAR), and classification pipelines for ecology research.

RESEARCH EXPERIENCE

University of Florida, Machine Learning and Sensing Lab **Aug. 2021 - Dec. 2026**
Doctoral Researcher *Gainesville, FL*

Self-Supervised Representation Learning for Hyperspectral Imagery (HSI)
- Developed a **self-supervised ViT** pretraining method that masks decomposed spectral components instead of raw channels, fixing trivial-or-impossible reconstruction targets; beats input-space masking on all 6 SpectralEarth segmentation tasks.
- Achieved 15% F1 score improvement over supervised baselines on tree species classification using self-supervised and semi-supervised learning with 2M unlabeled samples from NEON sites.

Pixel-Level Soil Water Content Prediction Using Deep Learning
- Developed first approach for pixel-level soil moisture mapping in rhizosphere systems using DenseNet-inspired CNN and hyperspectral imaging (R^2=0.89 at ~0.5mm resolution), with **UNet** segmentation for root-soil separation; led creation of the HyperPRI dataset.

PUBLICATIONS

- Graves, S. J., **Chowdhry, R.**, ... & Bohlman, S. A. (2025). Facilitating macrosystem biology with organismal-scale airborne remote sensing: Challenges and opportunities. Functional Ecology.
- Chang, S. J., **Chowdhry, R.**, Song, Y., ... & Zare, A. (2024). HyperPRI: A dataset of hyperspectral images for underground plant root study. Computers and Electronics in Agriculture, 225, 109307.